CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 07

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Weather forecasts are less reliable far into the future because: 1

(a) air has no temperature
(b) tiny differences in conditions grow over time
(c) instruments stop working after two days
(d) the Sun's heat stops changing

2. In earth science, the Earth is sometimes modelled as: 1

(a) a single point
(b) a hollow cube
(c) a smooth sphere layered into distinct regions
(d) a flat disc

3. 6 g of salt is dissolved in 34 g of water. The mass by mass percentage of the solution is: 1

(a) 15 % **(b)** 7.5 % **(c)** 25 % **(d)** 17.65 %

4. The solubility of gases in liquids generally: 1

(a) increases with temperature
(b) stays the same
(c) is zero at room temperature
(d) decreases with an increase in temperature

5. A molecule is best defined as: 1

(a) an electrically neutral entity of more than one atom capable of independent existence
(b) a charged particle of an element
(c) a single atom of any element
(d) only a particle of a compound

6. The molecular/formula unit mass of ammonia, NH_3 is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 17 u **(b)** 19 u **(c)** 34 u **(d)** 15 u

7. An atom of calcium has atomic number 20 and mass number 40. The number of neutrons in it is: 1

(a) 21 **(b)** 40 **(c)** 19 **(d)** 20

8. The velocity of a car changes from 0 m s^{-1} to 20 m s^{-1} in 5 s in a straight line. Its average acceleration is: 1

(a) 8 m s^{-2} **(b)** 4 m s^{-2} **(c)** −4 m s^{-2} **(d)** 5 m s^{-2}

9. Displacement is: 1

(a) always positive
(b) the shortest straight-line change in position with a direction
(c) always equal to distance
(d) the total path length covered

10. The net force needed to give a body of mass 50 kg an acceleration of 2 m s^{-2} is: 1

(a) 25 N **(b)** 100 N **(c)** 52 N **(d)** 200 N

11. Action and reaction forces do not cancel because they: 1

(a) act on different objects **(b)** act in the same direction
(c) are unequal **(d)** act at different times

12. Which of the following is an example of a body possessing only potential energy? 1

(a) a stretched bow before release **(b)** a flowing river
(c) a moving car **(d)** a rolling ball

13. Sound of frequency 170 Hz travels at 340 m s^{-1}. Its wavelength is: 1

(a) 0.5 m **(b)** 4 m **(c)** 57800 m **(d)** 2 m

14. Cell wall is absent in: 1

(a) animal cells **(b)** fungal cells **(c)** plant cells **(d)** bacterial cells

15. A potato piece placed in 20% salt solution will: 1

(a) swell **(b)** stay unchanged
(c) shrink as it loses water **(d)** dissolve

16. Tracheids and vessels are the conducting elements of: 1

(a) phloem **(b)** bone **(c)** xylem **(d)** epithelium

17. Bryophytes lack: 1

(a) vascular tissue **(b)** cell wall **(c)** chlorophyll **(d)** rhizoids

18. The male reproductive part of a flower is the: 1

(a) petal **(b)** stamen **(c)** sepal **(d)** pistil

19. The release of an egg from the ovary is called: 1

(a) implantation **(b)** fertilisation **(c)** ovulation **(d)** menstruation

20. The largest reservoir of nitrogen is the: 1

(a) atmosphere **(b)** plants **(c)** soil **(d)** oceans

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Estimate the number of breaths a person takes in a day if the rate at rest is 14 breaths per minute. 2

22. Give two reasons why ionic compounds have high melting points. 2

23. What does the slope of a position–time graph represent? 2

24. How does a rocket move upward? 2

25. Why are mitochondria called the powerhouses of the cell? 2

26. Name the two conducting tissues in plants and their functions. 2

27. What happens to the ovary and the ovule after fertilisation? 2

28. What is a biogeochemical cycle? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Explain sublimation and deposition using the example of camphor and sand. 3

30. Why are noble gases chemically unreactive? Give configurations of neon and argon. 3

31. Two bodies of masses m and 4m have the same KE. Find the ratio of their speeds. 3

32. Sound in steel travels at 5000 m s^{-1} and in air at 340 m s^{-1}. How much sooner does a sound travel through 340 m of steel than through air? 3

33. Why is cork formed in older stems? State its function. 3

34. What are protochordates? Give an example. 3

35. State the main postulates of Dalton's atomic theory (any four). 3

OR

12 g of carbon combines with 32 g of oxygen to form CO_2. How much CO_2 is formed from 2.4 g of carbon? Which law is used? 3

36. Why is insolation at the surface less than the solar constant? 3

OR

Explain land and sea breezes. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** Water is formed when hydrogen and oxygen combine. 5

(i) Give the electronic configurations of H and O.

(ii) Explain the bonding in H_2O.

(iii) Why does water have different properties from H_2 and O_2?

(iv) What is the mass ratio of H : O in water?

OR

Case study: Students react 4.0 g calcium carbonate with 2.92 g hydrochloric acid in a closed container and obtain 1.76 g CO_2, 0.72 g H_2O and $CaCl_2$. 5

(i) Find the mass of $CaCl_2$.

(ii) Which law is verified?

(iii) What would happen to the measured mass if the vessel were open?

(iv) Why?

38. **Case study:** A car of 800 kg moving at 25 m s^{-1} stops in 10 s. 5

(i) Find the retardation.

(ii) Find the braking force.

(iii) Find the stopping distance.

(iv) If the speed were doubled for the same force, find the new stopping distance.

OR

(i) Distinguish balanced and unbalanced forces. 5
(ii) Two teams pull a rope with 400 N and 300 N. Find the net force and direction.
(iii) Which team wins and why?

39. (i) How does a light microscope work? 5
(ii) Define resolution and contrast.
(iii) Find the magnification with 10X eyepiece and 100X objective.

OR

Case study: Potato pieces of equal weight are placed in water and in 20% salt solution for two hours. 5
(i) What happens to each?
(ii) Explain with osmosis.
(iii) Why were the pieces of equal size?
(iv) What would happen in an isotonic solution?

40. (i) Define reproduction. 5
(ii) Compare asexual and sexual reproduction in four points.
(iii) Why are clones formed?

OR

(i) Explain budding, spore formation and vegetative propagation with examples. 5
(ii) What is common among them?

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND